



ATTENDIFY — EMPLOYEE ATTENDANCE MANAGEMENT SYSTEM

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ATTENDIFY — EMPLOYEE ATTENDANCE MANAGEMENT SYSTEM

1. Introduction

1.1 Background

Employee attendance is an important part of workforce management. Traditional attendance methods, such as manual registers and spreadsheets, can make it difficult to maintain accurate records, track attendance history, and generate reports efficiently.

Attendify — Employee Attendance Management System is a web-based application developed to digitize employee attendance management. The system provides a centralized platform for managing employee records, recording daily attendance, monitoring attendance statistics, maintaining attendance history, and generating monthly reports.

The application is developed using the MERN stack, consisting of React, Node.js, Express.js, and MongoDB.

1.2 Purpose of the System

The purpose of Attendify is to provide an organized and efficient solution for managing employee information and attendance records. It reduces dependence on manual record keeping and allows authorized users to access attendance information through a centralized web interface.

1.3 Scope

The system covers:

- * User registration and authentication
- * Employee record management
- * Daily attendance management
- * Employee attendance history
- * Attendance statistics
- * Dashboard monitoring
- * Monthly attendance reports

- * Department-based filtering
 - * CSV report export
 - * Profile management
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2. Objectives

The main objectives of Attendify are:

- * To provide a centralized platform for employee management.
 - * To digitally record and manage daily employee attendance.
 - * To maintain employee attendance history.
 - * To provide attendance statistics through a dashboard.
 - * To generate monthly attendance reports.
 - * To allow reports to be filtered by department.
 - * To provide CSV export functionality.
 - * To implement secure authentication and protected application access.
 - * To reduce manual attendance record management.
 - * To improve the accessibility and organization of attendance data.
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3. System Overview

Attendify is a full-stack web application that allows authenticated users to manage employees and their attendance records.

The system consists of three major layers:

- 1. Frontend** — Provides the user interface and application navigation.
 - 2. Backend**— Handles business logic, authentication, validation, and API requests.
 - 3. Database**— Stores user, employee, and attendance information.
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4. Functional Requirements

4.1 User Authentication

The system shall allow users to:

- * Register an account.
- * Log in using their credentials.
- * Log out of the application.
- * View their profile.
- * Update their profile information.

4.2 Employee Management

The system shall allow authenticated users to:

- * Add new employees.
- * View employee records.
- * Edit employee information.
- * Delete employees.
- * Search employees.
- * Filter employees by department.
- * View employee attendance history.

4.3 Attendance Management

The system shall allow users to:

- * Mark employee attendance.
- * Update existing attendance records.
- * View today's attendance.
- * Identify employees whose attendance has not been marked.
- * Record time in and time out.
- * Add attendance notes.

4.4 Attendance History

The system shall provide individual attendance history for employees.

The history shall display attendance records and summary statistics such as:

- * Present days
- * Absent days
- * Late days
- * Leave days
- * Half days
- * Holiday days
- * Working days
- * Attendance percentage

4.5 Dashboard

The dashboard shall provide an overview of attendance information, including:

- * Total employees
- * Present employees
- * Absent employees
- * Late employees
- * Employees on leave
- * Half-day employees
- * Unmarked employees
- * Monthly attendance percentage
- * Recent attendance records

4.6 Monthly Reports

The system shall allow users to generate monthly attendance reports based on:

- * Month
- * Year
- * Department

4.7 CSV Export

The system shall allow users to export monthly attendance information as a CSV file. The exported report contains employee and department information along with attendance totals for different attendance statuses.

5. Non-Functional Requirements

5.1 Security

The system shall protect user accounts and application data through authentication, password hashing, protected routes, and HTTP-only cookies.

5.2 Usability

The interface should provide clear navigation, understandable forms, notifications, and responsive layouts.

5.3 Performance

The system should efficiently retrieve employee and attendance records using pagination, filtering, and optimized database queries where applicable.

5.4 Maintainability

The application is organized into separate frontend and backend modules to make the system easier to maintain and extend.

5.5 Data Integrity

The system validates important fields and prevents duplicate employee and attendance records where required.

6. Technology Stack

6.1 Frontend Technologies

- * **React**— User interface development
- * **Vite**— Frontend development and build tool
- * **React Router** — Application routing
- * **Axios**— API communication
- * **Tailwind CSS** — User interface styling

6.2 Backend Technologies

- * **Node.js** — Server-side runtime
- * **Express.js** — REST API development
- * **Mongoose** — MongoDB interaction
- * **JWT** — Authentication
- * **Bcryptjs** — Password hashing
- * **Cookie Parser** — Cookie handling
- * **CORS** — Cross-origin request handling

6.3 Database

MongoDB is used as the primary database for storing users, employees, and attendance records.

7. System Architecture

Attendify follows a client-server architecture.

7.1 Frontend Layer

The React frontend provides the graphical user interface through which users interact with the system. It handles navigation, forms, data presentation, validation, and API communication.

7.2 Backend Layer

The Node.js and Express.js backend handles:

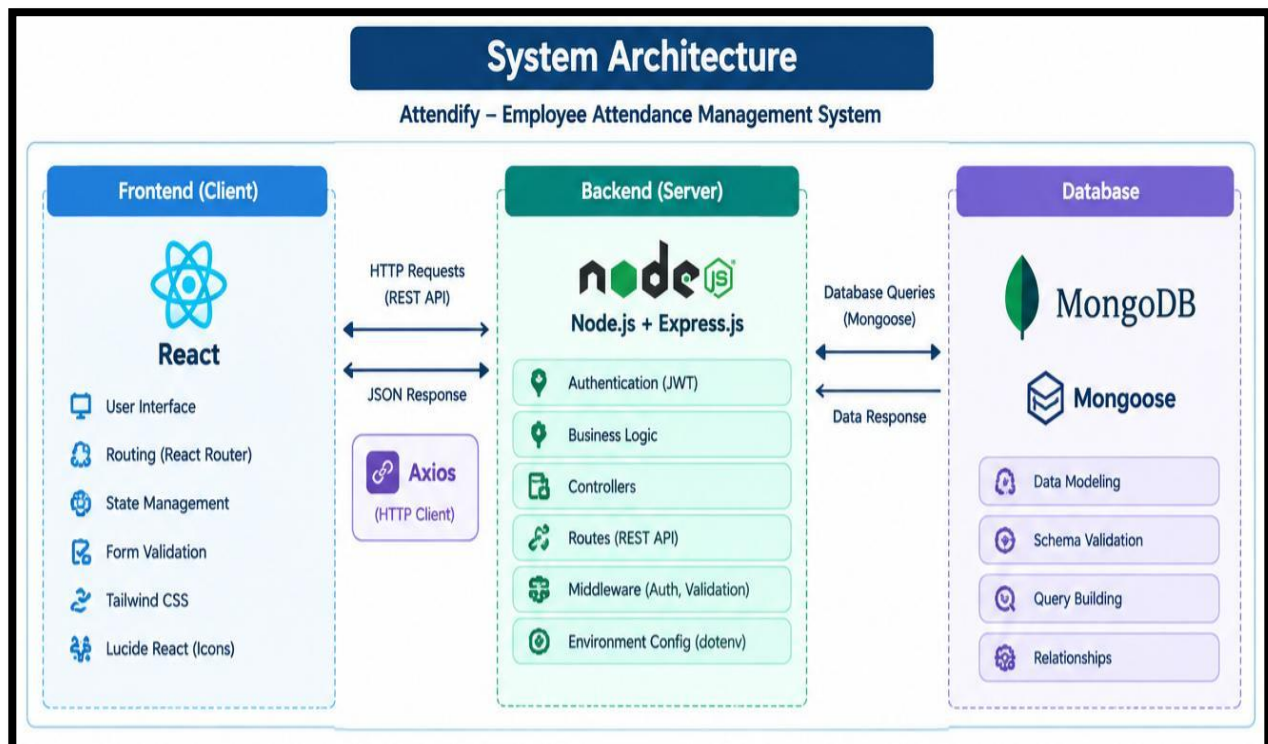
- * Authentication
- * Authorization
- * Business logic
- * Employee operations
- * Attendance operations
- * Dashboard statistics
- * Report generation
- * API requests

7.3 Database Layer

MongoDB stores the application's persistent data. Mongoose is used to define schemas and communicate with MongoDB.

7.4 Communication Flow

The frontend sends HTTP requests to the backend through Axios. Express routes process the requests and communicate with MongoDB through Mongoose. The resulting data is then returned to the frontend.



8. System Modules

8.1 Authentication Module

This module handles registration, login, logout, and profile management.

JWT tokens are stored in HTTP-only cookies to maintain authenticated sessions.

8.2 Employee Module

This module provides complete employee CRUD functionality.

Users can add, view, update, search, filter, and delete employee records.

8.3 Attendance Module

This module handles daily employee attendance. Users can mark and update attendance and view employees whose attendance has not yet been recorded.

8.4 Dashboard Module

The dashboard combines attendance information into summary statistics that provide a quick overview of workforce attendance.

8.5 Reports Module

The reports module generates monthly attendance reports based on selected month, year, and department.

8.6 Profile Module

The profile module allows authenticated users to view and update their basic profile information.

9. Database Design

Attendify uses three primary Mongoose models.

9.1 User Model

The User model stores:

- * Name
- * Email
- * Factory/Shop Name
- * Password
- * Created and updated timestamps

Passwords are hashed using bcryptjs before being stored.

9.2 Employee Model

The Employee model stores:

- * Full Name
- * Father's Name
- * Phone

- * CNIC
- * Department
- * Address
- * Joining Date
- * Status
- * Photo
- * Created and updated timestamps

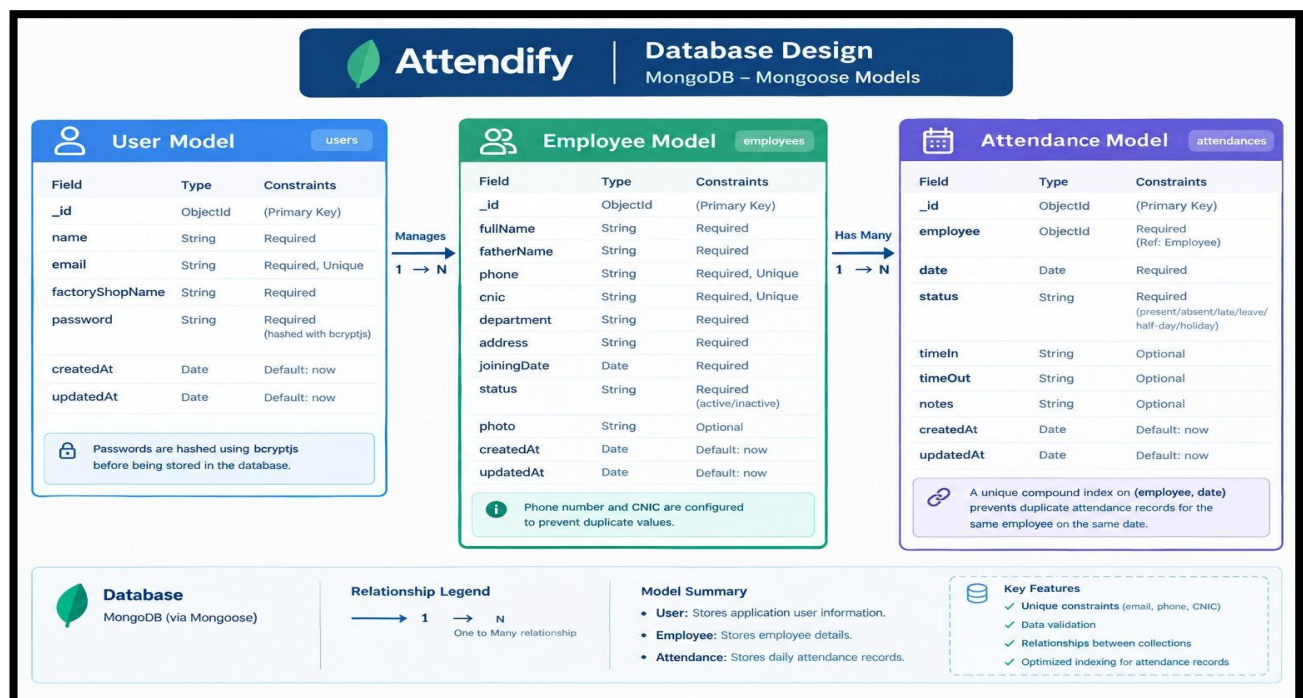
Phone number and CNIC are configured to prevent duplicate values.

9.3 Attendance Model

The Attendance model stores:

- * Employee reference
- * Date
- * Status
- * Time In
- * Time Out
- * Notes
- * Created and updated timestamps

The employee field references the Employee model.



10. Testing and Validation

The application can be validated through functional testing of its major modules.

Important test scenarios include:

- * User registration with valid and invalid data
- * Login with valid and invalid credentials
- * Employee creation and validation
- * Employee search and filtering
- * Employee update and deletion
- * Attendance marking
- * Duplicate attendance prevention
- * Attendance update
- * Attendance history calculation
- * Dashboard statistics
- * Monthly report filtering
- * CSV export
- * Protected route access

Testing ensures that the major system functions behave according to their requirements.

11. Advantages of the System

Attendify provides several advantages over manual attendance management:

- * Centralized employee information
 - * Faster attendance recording
 - * Easier attendance tracking
 - * Quick access to attendance history
 - * Automated attendance statistics
 - * Monthly reporting
 - * CSV data export
 - * Secure user authentication
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12. Conclusion

Attendify — Employee Attendance Management System provides a centralized digital solution for managing employee information and attendance records. The system supports secure authentication, employee CRUD operations, daily attendance tracking, attendance history, dashboard statistics, monthly reports, department filtering, and CSV export.

The project demonstrates the practical implementation of a full-stack web application using the MERN stack. The separation of frontend, backend, and database layers provides a structured architecture, while authentication, validation, and duplicate prevention mechanisms help maintain application security and data integrity.

Overall, Attendify provides an organized and scalable foundation for digital employee attendance management and can be further extended with features such as biometric attendance, payroll integration, notifications, and role-based access control.